

Apply a Screen Film to a Device SOP (1x Episode: 1 Device)

One episode films one whole device: the glass wiped, the backing peeled, the film landed on the glass, the bubbles rubbed out, one look under the lamp, and the device trayed. The table begins with one **device** (a phone) lying face up on the **work spot** in the center of the table, with its long edges running front to back and its **top end** at the back. There is no cradle: the device lies straight on the table and is held down by a gripper whenever a tool works on it. The **film tray**, with one **film** standing upright in it, starts in one of three places, set out below. The **cloth** lies flat on the **left spot**, just left of the device; once it has been used it goes into the film tray, in front of the film. The peeled **backing** is laid on the same left spot later. The **output tray** sits just right of the device. The **lamp** stands at the back-right and lights the **light spot** in the air in front of it.

The order never changes: wipe the glass, pick the film up, peel the backing, land the film, rub the bubbles out, then lift the device, hold it under the lamp, and tray it. No backing is peeled until the glass has been wiped, the cloth is in the film tray, and the device has been turned back square. The film is never landed with the backing still on it. No rubbing starts until the film is lying on the glass. The device is not lifted until no bubble can be seen, and the episode does not end until the device has been held under the lamp and set in the output tray.

The two grippers share the work, and the film changes hands several times. The **left gripper** lifts the cloth and passes it to the right gripper, pushes the device round a quarter turn anti-clockwise and holds it down while it is wiped, holds the film by its bottom edge while the peel is started, pulls the backing all the way off and lays it on the left spot, takes the film back by its bottom edge and lands it on the glass, holds the device down while the bubbles are rubbed out, and holds the top end of the device while the right gripper takes hold of it for the lift. The **right gripper** wipes the glass, drops the cloth in the film tray (in Config L it passes the cloth back to the left gripper, which drops it in), pushes the device back round a quarter turn clockwise to square, lifts the film out of the tray in Config M and R, starts the peel with one finger and holds the film by its top edge while the backing comes off, rubs out every bubble with its closed tip, lifts the device by its bottom end, holds it under the lamp, and sets it in the output tray.

The table is set up in one of three ways. Only the film tray, with the film in it, moves; the device, the left spot, the output tray, and the lamp are in the same place in all three.

- **Config L:** the film tray is at the back-left.
- **Config M:** the film tray is at the front-center, just in front of and a little left of the device.
- **Config R:** the film tray is at the right-center of the table, behind the output tray.

Where a step depends on the setup it says so on an **IF** line. Look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the film tray starts at the back-left (**Config L**), the front-center (**Config M**), or at the right-center behind the output tray (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the film tray's side lifts the film out: the **left gripper** in Config L, the **right gripper** in Config M and R. The same gripper drops the used cloth into that tray. No arm reaches across the table.
- **Peel hold:** whichever gripper lifts the film, the peel is always started with the **left gripper** holding the film by its **bottom edge** and the **right gripper** free. In Config M and R the right gripper lifts the film by its top edge and the left gripper takes the bottom edge: one pass. In Config L the left gripper lifts the film by its top edge, passes it to the right gripper, which holds the right edge, and then takes the bottom edge itself: two passes.
- **Pass:** at every pass the gripper that has the part holds it still, the other gripper closes on it, and only once the second gripper is closed does the first gripper open and lift clear.
- **Fixed roles:** in every config the peel, the landing, the rubbing, the lamp check, and the traying are the same. The right gripper always starts the peel and holds the film by its top edge while the left gripper pulls the backing, and the left gripper always lands the film.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **Close-up camera:** framed on the work spot, close enough that the film's edges, a bubble the size of a fingertip, and a lifted corner can be checked in review.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the device on the work spot, the film tray in the start zone for this episode's config, the cloth on the left spot, the output tray, and the lamp with the light spot in front of it.
3. The device is visible from above, so the whole **glass** and all four of its edges can be seen, both when it lies long edges front to back and when it has been turned to lie long edges left to right.
4. A closer view of the glass is recorded as well as the environment camera: the close-up camera is framed on the work spot so the film's edges against the glass, a bubble the size

of a fingertip, and a lifted corner can all be checked in review, and it also shows the device when it is held in the light spot.

5. Both arms are at home with grippers open.
6. The table is bare apart from the device, the film tray, the cloth, the output tray, the lamp, and robot hardware.
7. The right arm reaches the four corners of the glass, the film tray (Config M and R), the front-center where the cloth and the film are passed and the film is peeled, the light spot under the lamp, and the output tray without stretching. The left arm reaches the left spot, both ends of the device, the film tray (Config L), and the front-center without stretching.

Materials checklist

1. One **device** lies **face up** on the **work spot** in the center of the table, flat on the table and **square** to the front edge, with its long edges running front to back and its **top end** at the back. It slides on the table when a gripper pushes its side, and stays where it is put when the push stops.
2. The glass is bare: no film on it, no old glue, no tape.
3. The **film tray** sits flat and square to the front edge, in the place for this episode's config:
 - **Config L**: back-left.
 - **Config M**: front-center, just in front of and a little left of the device.
 - **Config R**: right-center of the table, behind the output tray.
4. One **film** stands **upright** in the film tray, leaning against the tray's back wall, with its **face** toward the front of the table and its **backing** toward the back wall. Its **top edge** is up. It stands in the tray, not on the table beside it. The film is cut to match the glass.
5. The backing has one **tab**, at the **top edge**, sticking up past the top of the film. The film itself has no tab: it is held by its edges.
6. The film tray is deep enough to hold the film upright, and has room on its floor in front of the film for the used cloth to be dropped in without touching the film.
7. One **cloth** lies flat on the **left spot**, just left of the device, folded to a pad about the size of a gripper tip, clean, dry, and not shedding fluff.
8. The **output tray** sits just right of the device, empty, flat, and square to the front edge. It is lined so a device set down in it lies flat.
9. The **lamp** stands at the back-right, switched on, with its head turned down and a little forward, so a device held level in the **light spot** in front of it has its glass lit from a low angle.
10. Keep the left of the device clear apart from the cloth on the left spot and, in Config L, the film tray at the back-left. The left gripper moves between the left spot, the device, the front-center, and in Config L the film tray.
11. Before collection, confirm by hand that:

- the device can be pushed round a quarter turn on the work spot by its sides without lifting, and stays put when the push stops;
- a film standing in the tray stays upright and does not fall over when the tray is nudged;
- a cloth dropped into the tray in front of the film lands without touching the film;
- a gripper finger pushed in beside the tab at the top edge lifts the backing away from the film without the film following it;
- once the peel is started, the rest of the backing pulls off the film in one steady pull by its tab;
- a film held only by one edge does not fold over on itself;
- a film laid on the glass can be nudged by its edges before it is pressed;
- a closed gripper tip rubbed lightly over a bubble pushes it out to an edge without lifting the film;
- a device held level in the light spot shows a bubble on the glass.

Workspace layout

- **Work spot:** the center of the table, where the device lies all episode until it is lifted
- **Start zone:** the film tray with one film: back-left (Config L), front-center just in front of and a little left of the device (Config M), or at the right-center behind the output tray (Config R)
- **Left spot:** just left of the device, where the cloth lies at the start and where the peeled backing is laid later
- **Front-center:** where the cloth is passed, where the film is passed, and where the film hangs while the backing comes off, in every config
- **Right of the device:** the output tray
- **Back-right:** the lamp, with the light spot in the air in front of it

Arm assignments

- **Left gripper:** lifts the cloth from the left spot and passes it to the right gripper; pushes the device round a quarter turn anti-clockwise and holds it down by its top end while the glass is wiped; in Config L takes the used cloth back and drops it in the back-left tray, then lifts the film out of that tray by its top edge and passes it to the right gripper; in every config takes the film by its bottom edge for the peel; pulls the backing all the way off by its tab and lays it on the left spot; takes the film back by its bottom edge and lands it on the glass; holds the device down by its left side while the bubbles are rubbed out; holds the top end of the device while the right gripper takes hold of the bottom end for the lift.
- **Right gripper:** takes the cloth from the left gripper, wipes the glass, and drops the cloth in the film tray (in Config L it passes the cloth back to the left gripper, which drops it in); pushes the device back round a quarter turn clockwise to square; in Config M and R lifts the film out of the tray by its top edge; in Config L holds the film by its right edge between the two passes; starts the peel with one finger, then holds the film by its top edge while the backing

comes off; rubs out every bubble with its closed tip; lifts the device by its bottom end, holds it in the light spot, and sets it in the output tray.

Vocabulary

- **Work spot:** the place in the center of the table where the device lies. It stays here until it is lifted in Step 6.
- **Square:** the long edges of a part run straight front to back, so neither end of the part sits nearer the left or the right than the other.
- **Device:** the phone being filmed. It lies face up all episode.
- **Top end:** the short end of the device that starts at the back. After the anti-clockwise turn it points left; after the clockwise turn it points to the back again.
- **Bottom end:** the short end of the device opposite the top end. It starts at the front.
- **Glass:** the flat shiny front face of the device that the film goes on.
- **Rim:** the thin flat border around the glass on the front of the device. A gripper presses here to hold the device down.
- **Push round:** a gripper presses against a side of the device and slides it across the table, then presses against another side, until the device has turned a quarter turn. The device is never lifted to turn it.
- **Cloth:** the folded cloth pad, one that does not shed fluff, used to wipe the glass. It is used once and then dropped in the film tray, in front of the standing film.
- **Left spot:** the patch of table just left of the device. The cloth lies here at the start; the peeled backing is laid here later.
- **Pass:** one gripper holds a part still, the other gripper closes on it, and only once the second gripper is closed does the first gripper open and lift clear. Every pass in this SOP happens a little above the table at the front-center.
- **Clean glass:** no dust specks, no lint, no smears, and no fingerprints anywhere on the glass.
- **Film:** the cut sheet that goes on the glass. Its **sticky side** goes down onto the glass; its **face** is the side that stays showing. The film has no tab and is held by its edges.
- **Crack:** a split, a snap line, or a sharp white crease in the film. A film with a crack is never landed.
- **Top edge, right edge, bottom edge of the film:** the edges as they are while the film stands in the tray and, once it is turned level, with the top edge toward the back of the table. The top edge is the one the backing's tab sticks out past.
- **Sticky side:** the underside of the film, the side the backing covers.
- **Backing:** the liner covering the sticky side of the film. It comes off in one piece and is laid on the left spot.
- **Tab:** the one pull tab, on the **backing**, sticking up past the film's top edge. The left gripper pulls it to take the backing off.

- **Peel started:** one finger of the right gripper has pushed in between the backing and the film at the top edge, so the backing has come away from the top of the film by about a finger's width and lies folded back under the film.
- **Film tray:** the tray the film stands in before it is picked up, and that the used cloth is dropped into. It starts at the back-left (**Config L**), the front-center (**Config M**), or at the right-center behind the output tray (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Standing:** the film is upright in the tray on its bottom edge, leaning on the back wall, face toward the front, top edge up.
- **Landed:** the film lies flat on the glass, sticky side down, let go by the gripper, and nothing has pressed it yet.
- **Even:** every edge of the film runs along the matching edge of the glass with the same narrow gap all the way round, and no corner of the film hangs over the rim.
- **Nudge:** a closed gripper tip pushes the edge of the landed film sideways a little without lifting it.
- **Bubble:** a pocket of air under the film. It shows as a bright or shadowed patch that moves when the gripper tip is rubbed over it.
- **Rubbed out:** the gripper tip has pushed the air in a bubble all the way past the edge of the film, so the patch is gone and not just moved to a new place.
- **Lamp:** the fixed lamp at the back-right. It is never moved and never touched by a gripper.
- **Light spot:** the place in the air in front of the lamp where a device held level has its glass lit from a low angle, so a bubble, a speck, or a lifted corner shows up.
- **Clear glass:** under the lamp, the film shows no bubble, no trapped speck, and no lifted corner.
- **Output tray:** the empty tray just right of the device that the finished device is set into.

Steps

Run Steps 1 to 6 in order on the one device, then end the episode with Step 7. Only Step 1.4 and Step 2 depend on where the film tray is. Every other line is the same in all three configs.

Step 1: Wipe the glass

Goal: the glass is clean edge to edge, the cloth is in the film tray, and the device lies square on the work spot again.

- The **right gripper** wipes with the **cloth** only.
- The **left gripper** holds the device down the whole time the cloth is on the glass.

1.1 Pass the cloth to the right gripper

- With the **left gripper**, pinch the **cloth** on the **left spot** and lift it straight up.

- With the **left gripper**, carry the cloth low over the table to a little above the **front-center**, in front of the device, and hold it still.
- With the **right gripper**, close on the cloth next to the left gripper.
- Open the **left gripper** and lift it clear. The cloth is now in the **right gripper** only.

1.2 Turn the device and hold it down

- With the **left gripper** closed, press against a side of the device, low down, and push it round on the table. Press against another side as needed, and keep pushing until the device has turned a **quarter turn anti-clockwise**, so its **top end** points left and its long edges run left to right.
- The **left gripper** never lifts the device to turn it. The device slides on the table the whole time.
- With the **left gripper**, press down on the **rim** at the **top end** of the device, now at its left end, and hold the device against the table.
- Keep the **left gripper** there through 1.3.

1.3 Wipe the glass

- With the **right gripper**, set the cloth flat on the glass and pass it across the screen.
- With the **right gripper**, keep passing the cloth across the glass, lifting and setting it down again as needed, until the glass looks clean. There is no set pattern and no set number of passes.
- With the **right gripper**, lift the cloth clear when the whole glass is clean.
- The **right gripper** never drags the cloth off the glass onto the table and back on, and never presses hard enough to move the device under the left gripper.

1.4 Drop the cloth in the film tray

The used cloth goes into the **film tray**, on the tray floor in front of the standing film. Look where the film tray is.

- Open the **left gripper** and lift it clear of the device.
- **IF the film tray is at the front-center (Config M) or at the right-center behind the output tray (Config R):** with the **right gripper**, carry the cloth low over the table to the **film tray** and let go over the tray floor in front of the film, without touching the film.
- **IF the film tray is at the back-left (Config L):** with the **right gripper**, carry the cloth to a little above the **front-center** and hold it still. With the **left gripper**, close on the cloth next to the right gripper. Open the **right gripper** and lift it clear (**pass**). With the **left gripper**, carry the cloth low over the table to the **film tray** at the back-left and let go over the tray floor in front of the film, without touching the film.

- The cloth goes in the **film tray** only, never in the output tray, on the device, or on the table. The film stays standing; the cloth does not knock it over or land on it.

1.5 Turn the device back

- With the **right gripper** closed, press against a side of the device, low down, and push it round on the table. Press against another side as needed, and keep pushing until the device has turned **a quarter turn clockwise**, so its **top end** points to the back again and it lies **square** on the **work spot**.
- The **right gripper** never lifts the device to turn it back.
- Lift the **right gripper** clear.

Check: the glass is **clean glass**: no dust specks, lint, smears, or fingerprints anywhere on it. Look before the cloth goes in the film tray. If any are left, wipe those places again with the same cloth before it goes in the tray. If the glass will not come clean, go on and log the device for a station check. After 1.5 the device lies square, long edges front to back, top end at the back, on the work spot.

Expected state: the glass is wiped, the cloth is in the film tray in front of the film, the device lies square on the work spot, both grippers are empty, and the film is still standing in the film tray with its backing on.

Step 2: Pick the film up and get it into the peel hold

Goal: the film hangs level from the left gripper by its **bottom edge**, face up, a little above the front-center, with its backing still on and the right gripper free.

- The film is held by its **edges**: top, right, or bottom.
- The film comes out of the tray **standing**, so it hangs upright as it is lifted. It is turned level and face up in the air, with its top edge toward the back of the table, before it is passed.
- Look at the film as it comes out. A film with a **crack** or a split in it is not used: lay it on the left spot with the gripper that holds it, go to Step 7, and log the device for a station check. The film is never bent hard enough to crack it while it is lifted, passed, peeled, or landed.

Look where the film tray is before reaching for the film.

- **IF the film tray is at the back-left (Config L):** with the **left gripper**, pinch the film across its **top edge** and lift it straight up out of the film tray, clear of the cloth, until its bottom edge is above the tray. Turn the **left gripper** so the film swings level, face up, with its top edge toward the back. Carry it level, low over the table, to a little above the **front-center** and hold it still. With the **right gripper**, close on the film's **right edge**. Open the **left gripper** and lift it clear (**first pass**). With the **left gripper**, close on the film's **bottom edge**. Open the **right gripper** and lift it clear (**second pass**).
- **IF the film tray is at the front-center (Config M) or at the right-center behind the output tray (Config R):** with the **right gripper**, pinch the film across its **top edge** and lift it straight

up out of the film tray, clear of the cloth, until its bottom edge is above the tray. Turn the **right gripper** so the film swings level, face up, with its top edge toward the back. Carry it level, low over the table, to a little above the **front-center** and hold it still. With the **left gripper**, close on the film's **bottom edge**. Open the **right gripper** and lift it clear (**pass**).

Then, in all three:

- Hold the film in the **left gripper** level and face up by its **bottom edge**, a little above the table at the **front-center**, with the **backing** still on it and the **tab** at the top edge pointing toward the back of the table.

Check: the film hangs level from the left gripper by its bottom edge and nowhere else, its face is up, its backing is whole, and the right gripper is empty and clear of it. Every pass had both grippers on the film for a moment before the first one opened. If the film swings or folds, hold still until it hangs flat before going on.

Expected state: the film is in the left gripper above the front-center by its bottom edge with its backing on, the film tray holds only the cloth, and the right gripper is empty.

Step 3: Peel the backing off the film

Goal: the film hangs from the left gripper by its bottom edge with its sticky side bare, and the backing lies on the left spot.

- The film is never set down between the peel and the landing.

3.1 Start the peel

- With the **left gripper**, keep holding the film still by its **bottom edge**, level and face up, above the front-center.
- With **one finger of the right gripper**, push in between the **backing** and the film at the **top edge**, beside the **tab**, and keep pushing until the backing has come away from the top of the film by about a finger's width and lies folded back under the film. This is the **peel started**.
- Push only this far. The **right gripper** does not pull the backing off with its finger, and does not lift the film.

3.2 Pull the backing off

- With the **right gripper**, keep its place and pinch the film across its **top edge**, on the strip the backing has come off. Hold the film still and level.
- Open the **left gripper** and let go of the bottom edge.
- With the **left gripper**, move to the top edge and pinch the **tab** of the backing, now folded back under the film.
- With the **left gripper**, pull the backing toward the **front** of the table, under the film, in one steady pull, keeping it low and flat, until the whole backing comes away from the film.

- Keep the **right gripper** still while the **left gripper** pulls. Do not pull upward, and do not jerk.
- If the backing tears or stops part way, pinch what is left of it with the **left gripper** and pull again the same way until none of it is on the film.

3.3 Take the film back

- With the **left gripper**, carry the backing low over the table to the **left spot**, just left of the device, lay it flat on the table, and let go.
- With the **left gripper**, come back to the film and pinch it across its **bottom edge**, so both grippers are on the film.
- Open the **right gripper**, let go of the top edge, and lift it clear.

Check: no backing is left on the film and the sticky side shows bare from edge to edge. The backing lies flat on the left spot, not in a tray, not on the device, and not still in the left gripper. The left gripper holds the film by its bottom edge and has not set it down. If the film folds over onto itself, do not try to peel it apart: go on to Step 4 only if the sticky side is still clean, otherwise finish the episode with the film as it is and log the device for a station check.

Expected state: the film hangs level from the left gripper by its bottom edge, sticky side down and bare, the backing lies on the left spot, the right gripper is empty, and the glass is still clean.

Step 4: Land the film on the glass

Goal: the film lies flat on the glass with its edges along the edges of the glass, and nothing has pressed it yet.

- Once the sticky side touches the glass the film is never lifted off again.
- With the **left gripper**, carry the film over the glass, holding it level and a little above it, with the film's **top edge** over the **top end** of the device and its bottom edge over the bottom end.
- With the **left gripper**, hold it there until each edge of the film looks in line with the matching edge of the glass. Nothing has touched the glass yet.
- With the **left gripper**, lower the film straight down, level, until its sticky side rests on the glass.
- Open the **left gripper** and let go of the bottom edge.
- Press nothing down. The film is only laid on.

Only if it is needed: if the film has landed with a wider gap on one side than the other, or a corner over the rim, both grippers **nudge** it: the closed tip of the **left gripper** on the left or top edge, the closed tip of the **right gripper** on the right or bottom edge, pushing that edge sideways a little at a time until the film is **even**. Do this as many times as it takes. Neither gripper lifts the film. A film that has landed even is left alone.

Check: the film is **landed**: flat on the glass, sticky side down, its edges along the edges of the glass, no corner over the rim, no corner standing up, and nothing has pressed on its face. Both grippers are empty and clear. If an edge will not nudge into line because the sticky side has grabbed the glass, leave the film where it is, go on to Step 5, and log the device for a station check.

Expected state: the film lies on the glass, both grippers are empty, the device is still square on the work spot, and no bubble has been rubbed out yet.

Step 5: Rub the bubbles out

Goal: no bubble can be seen under the film, with the device held down the whole time.

- The **left gripper** holds the device down while the **right gripper** rubs.
- The **right gripper** rubs with its **closed tip**, on the bubbles it can see, wherever they are. There is no set path and no set number of rubs.

5.1 Hold the device down

- With the **left gripper**, press down on the **rim** at the **left side** of the device and hold the device against the table.
- Keep the **left gripper** there through 5.2.

5.2 Rub each bubble out

- Close the **right gripper** fully.
- Look at the film. With the **right gripper**, set the closed tip down on the first **bubble**, press lightly, and rub it until the air goes out past an edge of the film. Rub in whichever direction works; it does not have to be the nearest edge.
- With the **right gripper**, do the same for every other bubble that can be seen, one at a time, wherever it is on the film. Go back to any bubble that has only moved, and rub it again until it is **rubbed out**.
- Keep going until no bubble can be seen. Press only hard enough to move the air. The **right gripper** never drags the tip off the film and back on, and never lifts a corner of the film to let air in.
- Lift the **right gripper** clear of the device.
- Open the **left gripper** and lift it clear of the device.

Check: no bubble can be seen on the film, every bubble that was worked on is **rubbed out** and not just moved, and no corner or edge of the film has lifted. The left gripper was on the rim the whole time the right gripper rubbed. Both grippers are clear of the device. If a bubble will not go, leave it: Step 6 shows it under the lamp and the device is logged.

Expected state: the film lies flat on the glass, the device lies square on the work spot, and both grippers are empty and clear.

Step 6: Lift the device, hold it under the lamp, and tray it

Goal: the device has been held level in the light spot for one look, and lies face up in the output tray with its film on.

- The lamp is never moved. The device goes to the light, not the light to the device.

6.1 Lift the device

- With the **left gripper**, press down on the **rim** at the **top end** of the device and hold it against the table.
- With the **right gripper**, pinch the device across its **bottom end**, low down, so it has a firm hold. The right gripper may touch the face of the film at the bottom end while it does this; it does not slide on it.
- Open the **left gripper** and lift it clear of the device.
- With the **right gripper**, lift the device straight up off the work spot, level and **face up**. Do not turn it over.

6.2 Hold it under the lamp and look

- With the **right gripper**, carry the device low over the table to the **light spot** in front of the lamp at the back-right.
- Hold the device still and level there, **face up**, so the lamp lights the glass from a low angle. Keep it still long enough to look over the whole glass once, from the top end to the bottom end, for bubbles, trapped specks, and lifted corners.
- Keep the arm to the side of the device, not over it, so nothing shades the glass or hides it from the camera.
- If the glass is **clear glass**, go on to 6.3. If a bubble, a speck, or a lifted corner shows, go on to 6.3 all the same and log the device for a station check. The device is not set back down and rubbed again once it has been lifted.

6.3 Set it in the output tray

- With the **right gripper**, carry the device low over the table to the **output tray**, just right of the work spot.
- With the **right gripper**, lower the device into the tray face up, **square** to the front edge, and let go.

Check: the device stopped in the light spot and was held still there, face up, with the arm to the side, while the whole glass was looked over. It lies flat in the output tray, face up, square to the front edge, with the film still on it and no corner lifted. If the device was carried past the lamp without stopping, that look does not count: lift it again and repeat 6.2. If the device sits crooked or on the tray lip, pick it up again by its bottom end with the **right gripper** and set it down flat.

Expected state: the device lies face up and square in the output tray with its film on, one look under the lamp has been made, the work spot is empty, and both grippers are empty.

Step 7: End the episode

Goal: recording ends with the device filmed, checked, and trayed.

1. Confirm the end state:
 - the film is on the glass, with no corner lifted;
 - the device was held under the lamp once;
 - the device lies face up and square in the output tray;
 - the work spot is empty;
 - the film tray holds the used cloth and nothing else, the backing lies on the left spot, and nothing else is on the table;
 - the lamp stands where it started.
2. Return both arms home with grippers open. Homing is the last thing the arms do.
3. Stop recording.

After the episode: reset the workspace

All reset work happens with recording off.

After each episode

1. Lift the device out of the output tray. Peel the film off the glass and throw it away with the backing from the left spot and the cloth from the film tray.
2. Clean the glass with a fresh cloth until no glue, dust, or smear is left on it.
3. Set the device face up on the **work spot** in the center of the table, **square** to the front edge, long edges front to back, top end at the back.
4. Stand one fresh **film** upright in the film tray, leaning on the back wall, face toward the front, top edge up, with the backing's **tab** sticking up past the top edge.
5. Lay one fresh **cloth** flat on the **left spot**, just left of the device, folded to a pad about the size of a gripper tip.
6. Set the film tray square in the place for the next episode's config: back-left (Config L), front-center (Config M), or at the right-center behind the output tray (Config R). Set the output tray square just right of the device. Check the lamp is on and pointing at the light spot.
7. Run both Setup checklists before the next episode.

At the end of the session

1. Inspect the parts. Throw away any used cloths. Check that the gripper tips are clean and dry, with no glue on them from the rubbing or the nudging.
2. Check the glass of the device under the lamp. If the glass is scratched, chipped, or will not come clean, pull that device out of collection.

3. Check that the lamp still lights and still points at the light spot. If it does not, flag the lamp and stop collection on this station.
4. Leave the device face up and square on the work spot with bare glass, one fresh film standing in the film tray with nothing else in it, one fresh cloth on the left spot, the output tray empty, and the lamp switched off.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain multiple violations; tag each separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table: the film tray is not in the start zone for the config; a gripper reaches across the table for the film or to drop the cloth; in Config L the right gripper reaches to the back-left for the film or with the cloth, or the film goes straight from the left gripper's top-edge hold to its bottom-edge hold without the right gripper holding the right edge in between; in Config M or R the left gripper lifts the film out of the tray itself, or the cloth is passed back to the left gripper; or the wrong IF line is followed.
- **SOP rule broken:** the start position, the same-side rule, and the peel hold (in Config M and R the right gripper lifts the film by its top edge and the left gripper takes the bottom edge; in Config L the left gripper lifts it, the right gripper holds the right edge, and the left gripper takes the bottom edge; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the film tray is before the first reach for the film, then follow that config's IF lines in Steps 1.4 and 2.

Violation: Cloth not passed from the left gripper

- **Visible cue:** the right gripper reaches to the left spot and picks the cloth up itself; the left gripper wipes the glass; the left gripper opens before the right gripper has closed on the cloth, so the cloth is dropped or thrown between the grippers; or the pass happens somewhere other than above the front-center.
- **SOP rule broken:** Step 1.1 (the left gripper lifts the cloth and holds it still above the front-center, the right gripper closes on it, and only then does the left gripper open).
- **Coaching note:** left lifts, right closes, then left opens. The cloth is never in the air on its own.

Violation: Device not turned, or turned the wrong way

- **Visible cue:** the glass is wiped with the device still lying long edges front to back; the left gripper turns the device clockwise, or by less or more than a quarter turn; the right gripper does the first turn or the left gripper does the second; a gripper lifts the device off the table to turn it instead of pushing it round; or the film goes on with the device still lying long edges left to right or out of square.
- **SOP rule broken:** Step 1.2 (the left gripper pushes the device round a quarter turn anti-clockwise before the cloth, without lifting it) and Step 1.5 (the right gripper pushes it round a quarter turn clockwise back to square after the cloth is in the film tray, without lifting it).
- **Coaching note:** left pushes it a quarter anti-clockwise to wipe, right pushes it a quarter clockwise to film. Push, never lift.

Violation: Device not held down

- **Visible cue:** the left gripper is off the rim while the cloth is on the glass or while the right gripper rubs, and the device slides, spins, lifts, or is chased across the table.
- **SOP rule broken:** Step 1.2 (the left gripper presses the top end of the device down the whole time the cloth is on the glass) and Step 5.1 (the left gripper presses the left side down the whole time the right gripper rubs).
- **Coaching note:** left gripper on the rim before the cloth or the tip touches the glass, and it stays there until they lift off.

Violation: Glass not wiped

- **Visible cue:** the cloth never leaves the left spot, or it never touches the glass, and the film goes on anyway.
- **SOP rule broken:** Step 1 (wipe the glass with the cloth before any backing is peeled).
- **Coaching note:** no film goes on unwiped glass. Wipe first, every time.

Violation: Dirty glass not re-wiped

- **Visible cue:** dust, lint, or a smear is still visible on the glass and the cloth goes into the film tray and the episode moves on.
- **SOP rule broken:** Step 1 Check (look before the cloth goes in the tray, and wipe any dirty place again with the same cloth).
- **Coaching note:** look before the cloth goes in the tray. Wipe the spot again, then move on.

Violation: Cloth not put in the film tray

- **Visible cue:** the cloth is dropped in the output tray, left lying on the table or on the device, dropped onto the standing film or knocks it over, or is still in a gripper while the device is turned back or the film is handled.
- **SOP rule broken:** Step 1.4 (drop the cloth on the film tray floor in front of the film before the device is turned back).
- **Coaching note:** cloth in the film tray, in front of the film, empty grippers, then turn the device.

Violation: Film set down before it is landed

- **Visible cue:** the film is put down on the table, in a tray, or on the device anywhere between the peel and the landing.
- **SOP rule broken:** Steps 3 and 4 (the film is not set down between the peel and the landing).
- **Coaching note:** once it is peeled it goes to the glass and nowhere else.

Violation: Film broken or cracked

- **Visible cue:** the film cracks, snaps, or gets a sharp crease while it is lifted, passed, peeled, nudged, or rubbed; or a film that already has a crack or a split is peeled and landed anyway.
- **SOP rule broken:** Steps 2 to 5 (a cracked film is laid on the left spot and logged, never landed; the film is held by its edges and never bent hard enough to crack it).
- **Coaching note:** look at the film as it comes out of the tray. Cracked means it goes on the left spot, not on the glass.

Violation: Peel hold not reached

- **Visible cue:** the peel is started with the film in the right gripper, or with the left gripper holding the top edge instead of the bottom edge; a gripper opens before the other gripper has closed on the film; or the film is passed somewhere other than above the front-center.
- **SOP rule broken:** Step 2 (the film ends up in the left gripper by its bottom edge with the right gripper free, through one pass in Config M and R and two passes in Config L, each above the front-center with the second gripper closed before the first one opens).
- **Coaching note:** left on the bottom edge, right free. Then the peel starts.

Violation: Peel started wrong

- **Visible cue:** the right gripper pulls the backing by its tab instead of pushing one finger in beside it; the finger goes in at an edge other than the top edge; the backing is pushed away by much more than a finger's width or pulled right off by the right gripper; the left gripper starts the peel; or the peel is started before the glass is wiped, before the cloth is in the film tray, or before the device is turned back.
- **SOP rule broken:** Step 3.1 (one finger of the right gripper goes in between the backing and the film at the top edge, beside the tab, about a finger's width and no more, only after Step 1 is complete).
- **Coaching note:** one finger in at the top, a finger's width, then pinch the top edge of the film.

Violation: Backing pulled wrong

- **Visible cue:** the left gripper pulls the tab upward, jerks it, or whips the backing across the film; the right gripper moves the film around while the left gripper pulls; the left gripper pulls the backing while it is still holding the film's bottom edge; or the film is landed with the backing, or a piece of it, still on the sticky side.
- **SOP rule broken:** Step 3.2 (with the film held still in the right gripper by its top edge, the left gripper lets go of the bottom edge, takes the tab, and pulls the backing toward the front in one steady low pull until none of it is left on the film).
- **Coaching note:** right holds the top still, left pulls the tab low and forward. One steady pull, all the way.

Violation: Backing not laid on the left spot

- **Visible cue:** the left gripper drops or lays the peeled backing in the film tray, in the output tray, on the device, anywhere on the table but the left spot, or on the floor; or it is still in the left gripper when the left gripper comes back for the film.
- **SOP rule broken:** Step 3.3 (carry the backing to the left spot, just left of the device, and lay it flat on the table).
- **Coaching note:** the backing goes flat on the left spot before the left gripper comes back for the film.

Violation: Film landed by the wrong gripper

- **Visible cue:** the right gripper carries the film over the glass and lays it down itself, without the left gripper taking the bottom edge back first; or the right gripper opens before the left gripper has closed on the bottom edge.
- **SOP rule broken:** Steps 3.3 and 4 (the left gripper takes the film back by its bottom edge, then the right gripper lets go, then the left gripper carries the film over the glass and lands it).
- **Coaching note:** right peels, left lands. Left takes the bottom edge back before the film goes over the glass.

Violation: Film not landed flat

- **Visible cue:** the left gripper drops the film onto the glass, lands one corner or one edge first, lets the film fall out of the gripper, lets go while the film is still in the air, or lands it with its top edge at the bottom end of the device.
- **SOP rule broken:** Step 4 (line every edge up in the air with the top edge over the top end, then lower the film straight down and level until its sticky side rests on the glass, then let go).
- **Coaching note:** line it up, lower it flat, then open. The film settles, it does not drop.

Violation: Film left crooked

- **Visible cue:** the landed film sits crooked on the glass, with a much wider gap on one side than the other or a corner over the rim, and the rubbing starts anyway with no nudge; or a gripper nudges the film by its face instead of its edge.
- **SOP rule broken:** Step 4 (a film that has landed crooked is nudged by its edges until it is even before any rubbing).
- **Coaching note:** look at all four edges before you rub. If it is crooked, nudge it even first.

Violation: Film lifted off the glass

- **Visible cue:** after the film has touched the glass, either gripper peels it back up, lifts a corner, or takes it off and lays it down again.
- **SOP rule broken:** Step 4 (once the sticky side touches the glass the film is nudged if it needs it, never lifted; a film that will not come even is left as it is and logged).
- **Coaching note:** nudge, never lift. A film that will not come even gets logged, not re-laid.

Violation: Bubble left on the film

- **Visible cue:** a bubble can still be seen on the film when the right gripper lifts clear and the device is lifted; a rub pushes a bubble to a new place on the film instead of past an edge and it is not rubbed again; or the right gripper rubs with its tip open, or drags the tip off the film and back on.
- **SOP rule broken:** Step 5.2 (with the tip closed, rub each bubble that can be seen until the air goes out past an edge, and keep going until no bubble can be seen).
- **Coaching note:** tip closed, find every bubble, push it out past the edge. Look again before you lift.

Violation: Lamp check skipped or done the wrong way

- **Visible cue:** the device goes from the work spot straight to the output tray without stopping in the light spot; it is carried past the lamp without being held still, or held face down; the arm stands over the glass through the whole look, shading it or hiding it from the camera; or a gripper touches or moves the lamp.

- **SOP rule broken:** Step 6.2 (carry the device to the light spot, hold it still and level, face up, with the arm to the side, long enough to look over the whole glass once, and never move the lamp).
- **Coaching note:** every device gets one look under the lamp. Level, face up, arm to the side, stop and look.

Violation: Device lifted or trayed wrong

- **Visible cue:** the right gripper takes hold of the device with the left gripper not holding the top end; the right gripper takes hold anywhere but the bottom end; the device is turned over or tilted steeply while it is carried; or it is left face down, crooked, or on the tray lip in the output tray.
- **SOP rule broken:** Steps 6.1 and 6.3 (the left gripper holds the top end while the right gripper pinches the bottom end, then the device is carried level and face up and set flat and square in the output tray).
- **Coaching note:** left holds the top, right takes the bottom. Level, face up, square in the tray.

Violation: Work done out of order

- **Visible cue:** the phases run out of sequence: the film is picked up before the cloth is in the film tray or before the device is turned back, the backing is peeled before the glass is wiped, the film is landed before the backing is off, the rubbing starts before the film is on the glass, or the device is lifted while a bubble can still be seen and is not being worked on.
- **SOP rule broken:** Steps 1 to 6 (wipe, pick up, peel, land, rub, then lift, lamp, and tray, in that order).
- **Coaching note:** finish the phase before you start the next one. Say the phase name before you move.

Violation: Wrong arm used

- **Visible cue:** an action assigned to one gripper is done by the other, including the right gripper lifting the cloth from the left spot, doing the anti-clockwise turn, pulling the backing all the way off, or landing the film; or the left gripper wiping, doing the clockwise turn, starting the peel with a finger, rubbing bubbles, carrying the device, or setting it in the output tray.
- **SOP rule broken:** Steps 1 to 6 (the left gripper passes the cloth, turns the device anti-clockwise and holds it, holds the film's bottom edge for the peel, pulls the backing, lands the film, holds the device for the rubbing, and holds its top end for the lift; the right gripper wipes, turns the device clockwise, starts the peel and holds the film's top edge, rubs, and lifts, lamps, and trays the device).
- **Coaching note:** left holds, peels, and lands; right wipes, starts the peel, rubs, and carries. It does not swap.

Violation: Part dropped or knocked off its spot

- **Visible cue:** either gripper drops the device, the film, the cloth, or the backing on the glass, the table, or the floor, knocks the standing film over in its tray, or tips, pushes, or knocks the film tray, the output tray, or the lamp off its spot.
- **SOP rule broken:** Steps 1 to 6 (keep the device, the film, the cloth, the backing, and every tray and the lamp on their spots through the whole episode).
- **Coaching note:** work slower and lower over the table, and keep each carry short.

Violation: Wrong episode ending

- **Visible cue:** recording stops before the film, the lamp look, and the trayed device are confirmed, an arm is not home, a gripper is closed, or an arm does something else after homing.
- **SOP rule broken:** Step 7 (confirm the end state, return both arms home with grippers open as their final action, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Non-violation failures

Failures not caused by how the task was run are system issues. Log and discard the episode rather than tagging them as SOP violations.

- Recording stops or pauses during the episode.
- A camera drops frames or loses its feed.
- An arm or gripper fails, drifts, or reports a motor error.
- The film arrives cut to the wrong size, so it cannot lie along the edges of the glass.
- The film arrives creased, curled, or with its sticky side already dried out, so a correct landing still leaves it lifted.
- The backing arrives already torn or stuck fast to the film, so a finger cannot start the peel or a steady pull cannot take it off in one piece.
- The tab is missing or falls off the backing before it is used.
- The device glass is chipped or scratched under the film area before the episode starts.
- The device will not stay put on the work spot when the push stops, however carefully it is pushed.
- The cloth sheds lint onto the glass no matter how it is wiped.
- The lamp fails to light, or its head will not stay pointed at the light spot.

Annotation subtasks (from SOP)

1. Lift the cloth with the left gripper and pass it to the right gripper above the front-center
2. Push the device round a quarter turn anti-clockwise with the left gripper and hold it down by its top end

3. Wipe the whole glass with the right gripper
4. Drop the cloth in the film tray in front of the film (right gripper in Config M and R; passed back to the left gripper in Config L)
5. Push the device round a quarter turn clockwise back to square with the right gripper
6. Lift the standing film out of the film tray by its top edge and turn it level (left gripper in Config L, right gripper in Config M and R)
7. Pass the film until the left gripper holds its bottom edge (two passes in Config L, one in Config M and R)
8. Start the peel with one finger of the right gripper at the top edge
9. Pinch the top edge of the film with the right gripper
10. Pull the backing all the way off by its tab with the left gripper and lay it on the left spot
11. Take the film back by its bottom edge with the left gripper
12. Land the film flat on the glass with the left gripper, and nudge it even only if it needs it
13. Hold the device down with the left gripper and rub out every bubble with the right gripper tip
14. Hold the top end with the left gripper, pinch the bottom end with the right gripper, and lift the device
15. Hold the device still in the light spot and look over the whole glass
16. Set the device face up and square in the output tray
17. Confirm the end state, return both arms home, and end the episode